NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Roll No. 124102019

Month and year: **Dec.'2024**
Program: **B.Tech. [CS, IT, MnC, AIDS, AIML]**
Subject: **Problem Solving & Programming Skills I**
Maximum Marks: **50**

Total no. of pages used: **2**
Semester: **1st**
Course code: **CSIC-101**
Duration: **03 Hrs**

NOTE:

1. ***Q.1 is compulsory.***
2. *Attempt any FIVE questions from rest of the questions (each of 6 marks).*
Thus, ***total attempted questions are SIX.***
3. *Assume suitable data, if missing. While writing a C program, consider the following facts:*
   (i) Use appropriate names for all Identifiers.
   (ii) Show all commands for compiling and running your program.
   (iii) Show the entered input(s) & expected output(s).
   (iv) Make use of Indentation properly.

Q1. (a) Find the output of the following C programs, if there is any compile time or runtime error then write the reason for the same. **[Marks: 10+5+5 = 20]**

(a)
```
#include<stdio.h>
int main()
{
  printf("%x", -1<<1);
  return 0;
}
```

(b)
```
#include<stdio.h>
int main() {
  int n;
  for(n = 7; n!=0; n--)
    printf("n = %d", n--);
  return 0;
}
```

(c)
```
#include <stdio.h>
int main()
{
  int a = 5;
  double b;
  printf("%d ", sizeof(++a + b));
  printf("%d ", a);
  return 0;        }
```

(d)
```
#include <stdio.h>
int main()
{
  int a = 5;
  int b = 10;
  {
    int a = 2;
    a++;
    b++;
  }
  printf("%d %d", a, b);
  return 0;
}
```

(e)
```
# include <stdio.h>
int main()      {
  int i=0;
  for(i=0; i<20; i++)
  {
    switch(i)
    {
      case 0:
        i+=5;
      case 1:
        i+=2;
      case 5:
        i+=5;
      default:
        i+=4;
        break;
    }
    printf("%d ", i);
  } return 0;
}
```

(b) Write a menu-driven C program using a switch and a do-while loop, which has the following options:

1. Take input for a number and calculate the factorial of a number.
2. Take input for a number and check whether the entered number is prime or not.
3. Exit

Note: Apply the do-while condition so that the Loop will continue printing the menu until the user enters number 3.

(c) Write a C program to display the following pattern for n rows using a number that starts with 1 and each row will have a 1 as the first and last number.

```
1
121
12321
1234321
```

Q2. Write a C program that will ask the user to enter the first 10 natural numbers and store these numbers in a 1-D array. Further, define a function **sumArray()** that will take an array of integers as an input and return the sum of elements of the array. Call this function from the **main()** function to find the sum of the first 10 natural numbers.

Q3. Write a C program that will take the following **string** as input from the user and count the number of spaces in the string: - "nit kkr haryana india".
Note that you must define a separate function, other than the main() function, named ***countSpaces()*** with appropriate parameters & return type.

Q4. Write a C program that will perform the following tasks:

i) Declare and initialize an integer variable **x** with value **5**.
ii) create two pointer variables that point to the variable **x**.
iii) change **x**'s value to **10** by one pointer through dereferencing and print the updated value of **x** by another pointer.

Q5. Write a C program that will dynamically allocate space for an integer and initialize(or assign) it to the value **10**. Reclaim this dynamically allocated space for the integer. Write the prototype (declaration) of the ***realloc()*** function in the C programming language.

Q6. Create a new composite(or aggregate) data type that consists of two members: an integer and a float by using the keyword 'struct'. Write a C program that will illustrate the usage of the **right arrow selection (->) operator** on this composite data type. Further, compare the memory requirements if this composite data type is defined using the keyword 'union'.

OR

Q6. Design a structure named "Car" to store details like car ID, model, and rental rate per day. Write a C program to input data for three cars, calculate the total rental cost for a specified number of days, and display the results.

Q7. Write a C program to take input from the user and check whether the given input is a valid identifier or not.

Q8. Write a C program to find the 'Perfect' numbers within a given number of ranges. A perfect number is a positive integer that is equal to the sum of its positive divisors.